

SIMATS ENGINEERING

ASSESSMENT TOOL 3 : Technical Presentation

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Imitation (Level 1)

Question Count: 5

Total Marks: 50

Code of Conduct

I BAIRAGANI DINESH(192372189)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____B DINESH_____



BIG DATA ANALYTICS IN CLOUD COMPUTING

Cloud-Based Data Processing Architectures, Analytics Platforms & Business Value



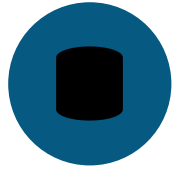
OBJECTIVE

Explore cloud-based data processing architectures, analytics platforms, business value, and real-world applications of Big Data Analytics.

Core Technical Explanation

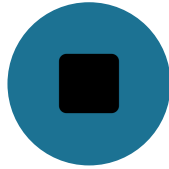
What Big Data Analytics in the Cloud Means

Big Data Analytics in Cloud Computing combines massive, fast, and varied datasets (Volume, Velocity, Variety) with elastic, on-demand cloud infrastructure — enabling organizations to store, process, and derive insight from data without owning physical hardware.



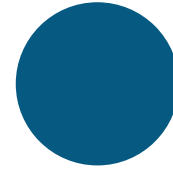
Storage Layer

Scalable object storage & data lakes (e.g., S3, ADLS) hold raw, structured, and unstructured data.



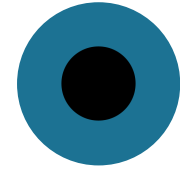
Processing Layer

Distributed engines (Hadoop, Apache Spark) run batch and stream computations across clusters.



Analytics & ML Layer

Managed platforms deliver querying, BI dashboards, and machine-learning driven insights.

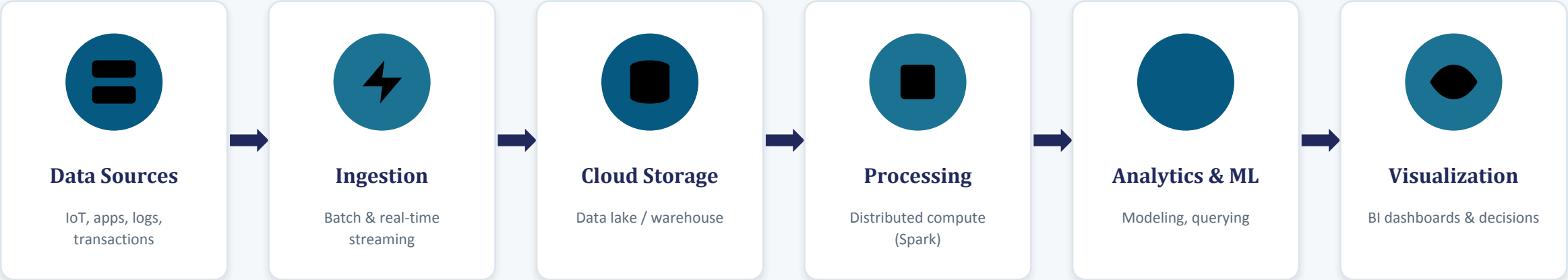


Elastic Delivery

Pay-as-you-go, auto-scaling resources provision instantly, worldwide, on demand.

Cloud Big Data Analytics Architecture

End-to-End Data Flow: Source to Decision



Elastic Scaling

Resources grow or shrink automatically with workload demand.

Distributed Processing

Data is partitioned and computed in parallel across nodes.

Managed Services

Cloud providers handle infrastructure, patching, and availability.

Real-World Use Case

Cloud-Powered Personalization at Global Scale



Video Streaming Platform

A global streaming service processes viewing behavior from 250M+ subscribers using a fully cloud-hosted big-data pipeline to power its recommendation engine.

Business Impact

Higher engagement, reduced churn, and content decisions driven directly by cloud-scale analytics.



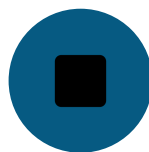
1. Real-Time Ingestion

Streaming events (clicks, pauses, watch time) are captured continuously via cloud pipelines.



2. Elastic Data Lake

Petabytes of viewing history are stored cost-effectively in cloud object storage.



3. Distributed Processing

Cluster-computing jobs analyze patterns across millions of users in parallel.



4. Personalized Output

ML models generate tailored recommendations delivered back to each user instantly.

Conclusion & Key Takeaway



Cloud computing transforms Big Data Analytics from a costly, hardware-bound effort into a scalable, on-demand capability accessible to organizations of any size.

Elastic Scale

Compute and storage grow with data volume — no upfront hardware investment.

Faster Insight

Distributed processing turns raw data into decisions in near real time.

Broad Access

Managed cloud platforms put advanced analytics within reach of any business.

KEY TAKEAWAY

The convergence of Big Data and Cloud Computing gives organizations the agility to turn massive, fast-moving data into real-time, evidence-based business decisions.

Limitations & Future Scope

Current Challenges and Where the Field Is Headed



LIMITATIONS

- Data privacy & security risks in multi-tenant cloud environments
- Vendor lock-in limits portability between cloud providers
- Unpredictable costs from large-scale storage & compute usage
- Network latency can hinder true real-time analytics
- Shortage of skilled cloud & big-data professionals



FUTURE SCOPE

- Edge-cloud integration for faster, localized processing
- AI-driven automated analytics & self-optimizing pipelines
- Serverless data platforms reducing operational overhead
- Expanded real-time streaming at massive scale
- Stronger built-in governance, privacy & compliance tooling